Shreeja Bandi

ai23btech11026@iith.ac.in | shreejanaidu7@gmail.com | +91 8179104857 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.45
Narayana Junior college	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2023	98.40%
Narayana School	Class X (Telanagana State Board of Secondary Education)	2021	10.00%

WORK EXPERIENCE

Tech Mahindra - TechnoRise Intern, Makers Lab | Pune May 2026 - July 2026

- Engineered an autonomous **confidence-weighted agent** for the **ARC-AGI-3 benchmark** using **grid-diff feedback loops** to dynamically adapt action probability distributions.
- Designed an **MD5-based state fingerprinting protocol** mapping action confidence to board states, enabling **zero-shot adaptive reasoning** without LLM inference.
- Built **'VectorHunter'** via the *arcengine* framework, resolving critical **spatial rendering, portal teleportation, gravity collision,** and win-state edge cases.
- Architected a **4-tier memory system** (current, cross-level, meta-data, world-model) and submitted the agent independently to **Kaggle's ARC-Prize 2026**.

PROJECTS

Enterprise IT Support Agentic RAG Copilot | Python, LangGraph, FastAPI, Pinecone Vector DB, Groq LLM, Tavily, Docker, HTML/CSS/JS

- Architected an **agentic Forward Deployed Engineer (FDE) copilot** using **LangGraph** to automate enterprise IT retrieval and eliminate search hallucinations.
- Implemented a stateful decision workflow featuring **intent routing, Pinecone vector search, multi-step evidence grading, Tavily web fallback,** and **dynamic query rewriting**.
- Engineered asynchronous **FastAPI** endpoints, **SQLite audit logging**, and an **RBAC-protected document ingestion pipeline** (.pdf, .md, .docx) wrapped in a **Dockerized UI**.

Robust Zero-Shot Coreset Selection via Density Sampling | PyTorch, DINOv2, Scikit-learn, k-NN Filtering, Distance Analysis

- Designed a **density-aware coreset selection pipeline** combining **k-NN noise filtering** with **DINOv2 embeddings** and **z-score distance metrics** to isolate high-value data sub-spaces.
- Achieved **93.44% accuracy on CIFAR-10 using only 5% of training data**, outperforming the ZCore baseline by **44% under high-noise conditions**.

Full-Stack Retail Revenue & Dynamic Inventory Optimizer | React.js, Node.js, Express.js, PostgreSQL, SQL Triggers, REST APIs

- Architected a **digital transformation (DX) prototype** for retail management, integrating **PL/pgSQL stored procedures and triggers** for real-time dynamic pricing.
- Automated **sales-driven price elasticity adjustments**, inventory stockout alerts, and **role-based access control (RBAC)** for admin and customer workflows.

Duplicate Question Pair Detection Pipeline | Python, NLP, NLTK, XGBoost, Random Forest, Scikit-learn

- Built an NLP classification pipeline engineering **22 similarity features** (token overlap, fuzzy matching) combined with **6,000 BOW features** across **30,000 question pairs**.
- Evaluated ensemble learning architectures, achieving a peak classification accuracy of **79.27% using XGBoost**.

SKILLS

Generative AI & Agentic Workflows: Agentic RAG, Multi-Agent Systems, LangGraph, Vector DBs (Pinecone, ChromaDB), LangChain, Prompt Engineering.

Machine Learning, DL & NLP: PyTorch, TensorFlow, Scikit-Learn, Transformers, BERT, CNNs, Vision Transformers (ViT), YOLO, Reinforcement Learning, Text Classification.

Languages & Web Engineering: Python, C, C++, SQL, JavaScript, HTML/CSS, React.js, Node.js, Express.js, PostgreSQL, MySQL, FastAPI.

Developer Tools & Cloud: Git, GitHub, Docker, VS Code, Jupyter Notebooks.

Project & Financial Coordination: System Design, Technical Planning, Event Budgeting & Cost Tracking, Iterative Development

RELEVANT COURSES

Artificial Intelligence & Core ML: Foundation of Machine Learning, Deep Learning, Agentic AI & Multi-Agent Systems, Natural Language Processing, Computer Vision, AI in Finance, Reinforcement Learning, Robotics.

Computer Science & Systems: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Architecture, Discrete Mathematics, Compilers, Game Theory.

Mathematical Foundations: Matrix Theory, Linear Systems & Signal Processing, Probability & Random Processes, Statistics, Numerical Analysis, Data Science Analysis.

POSITIONS OF RESPONSIBILITY

Finance Coordinator, Sunshine Management Team (2024-2025): Directed annual budget planning and financial operations for all campus events, managing cash advance disbursements, expenditure tracking, and post-event bill reconciliations.

Hospitality Coordinator, Diesta & Elan & nVision (2023-2025): Managed guest accommodations and logistics execution for major institute festivals, leading team members under high operational pressure.

EXTRACURRICULAR

National Cadet Corps (NCC): Cadette Member -- developed rigorous team discipline, situational leadership, and cross-functional adaptability through structured military drill training and field exercises.

LANGUAGES

Korean: Elementary Proficiency -- Fluent in Hangul (reading/writing); basic listening and conversational comprehension.

Japanese: Elementary / Beginner -- Actively studying JLPT N5 material and basic conversational phrases.

English: Native / Full Professional Proficiency